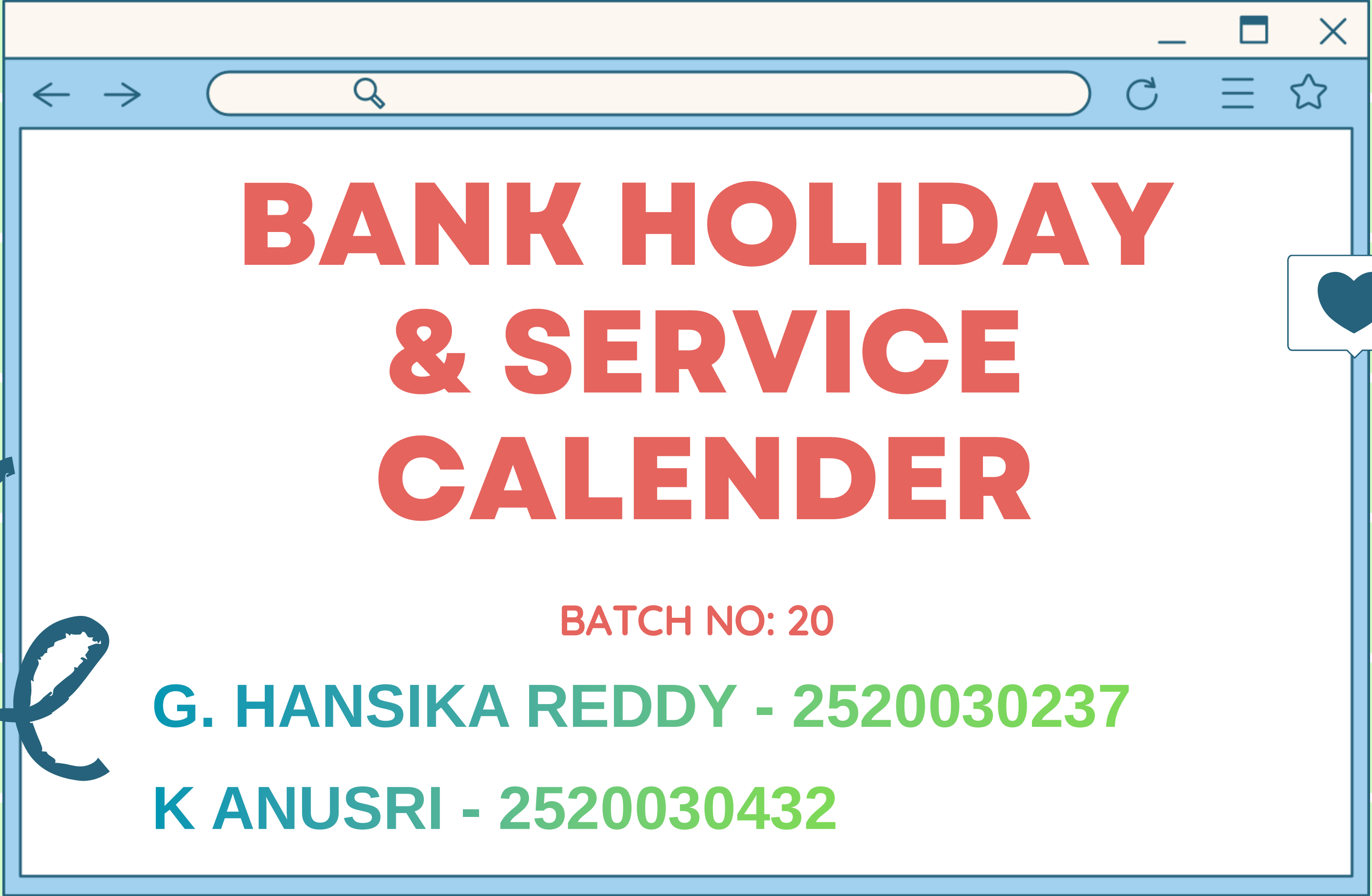


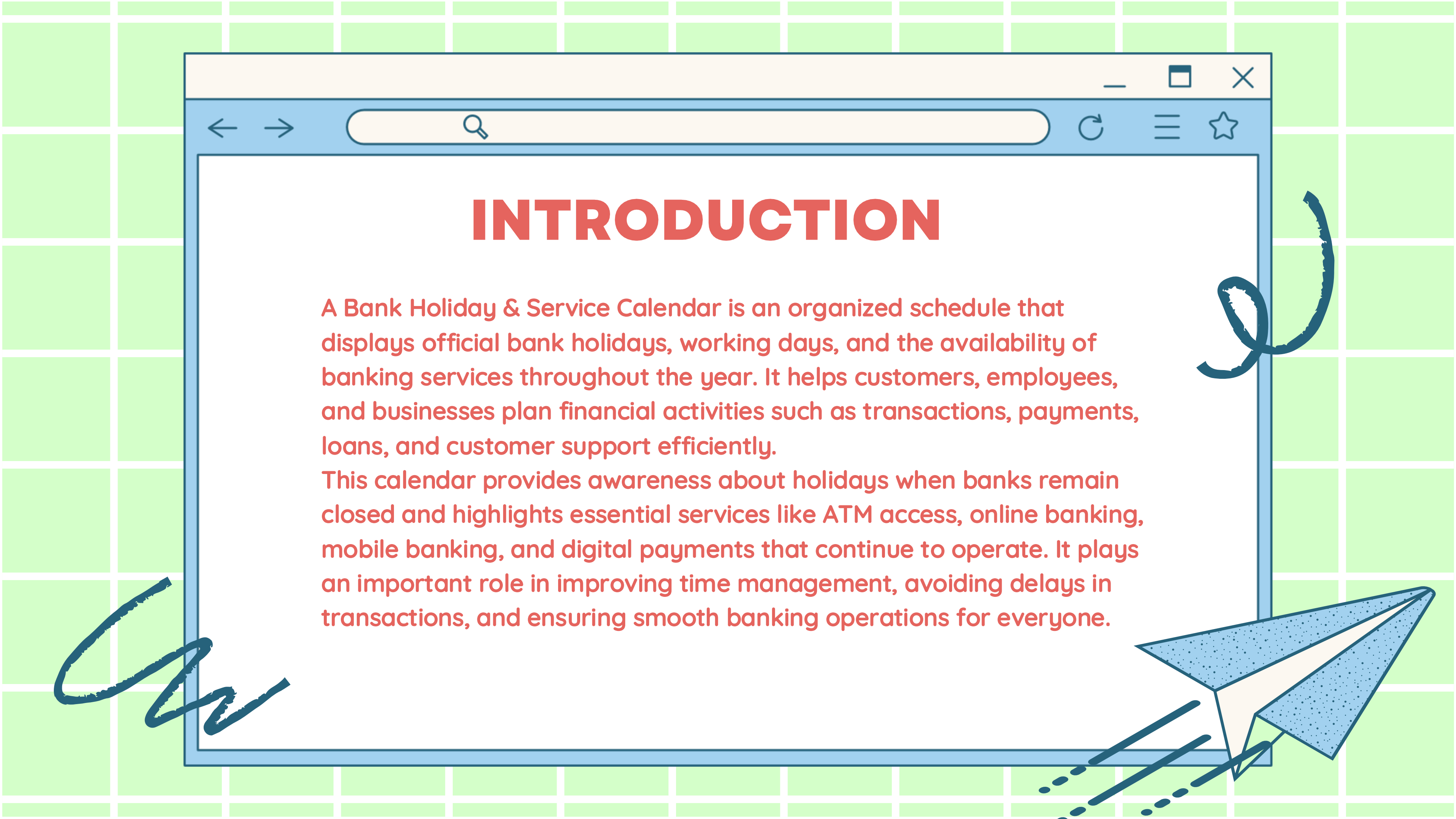


BANK HOLIDAY & SERVICE CALENDER

BATCH NO: 20

G. HANSIKA REDDY - 2520030237

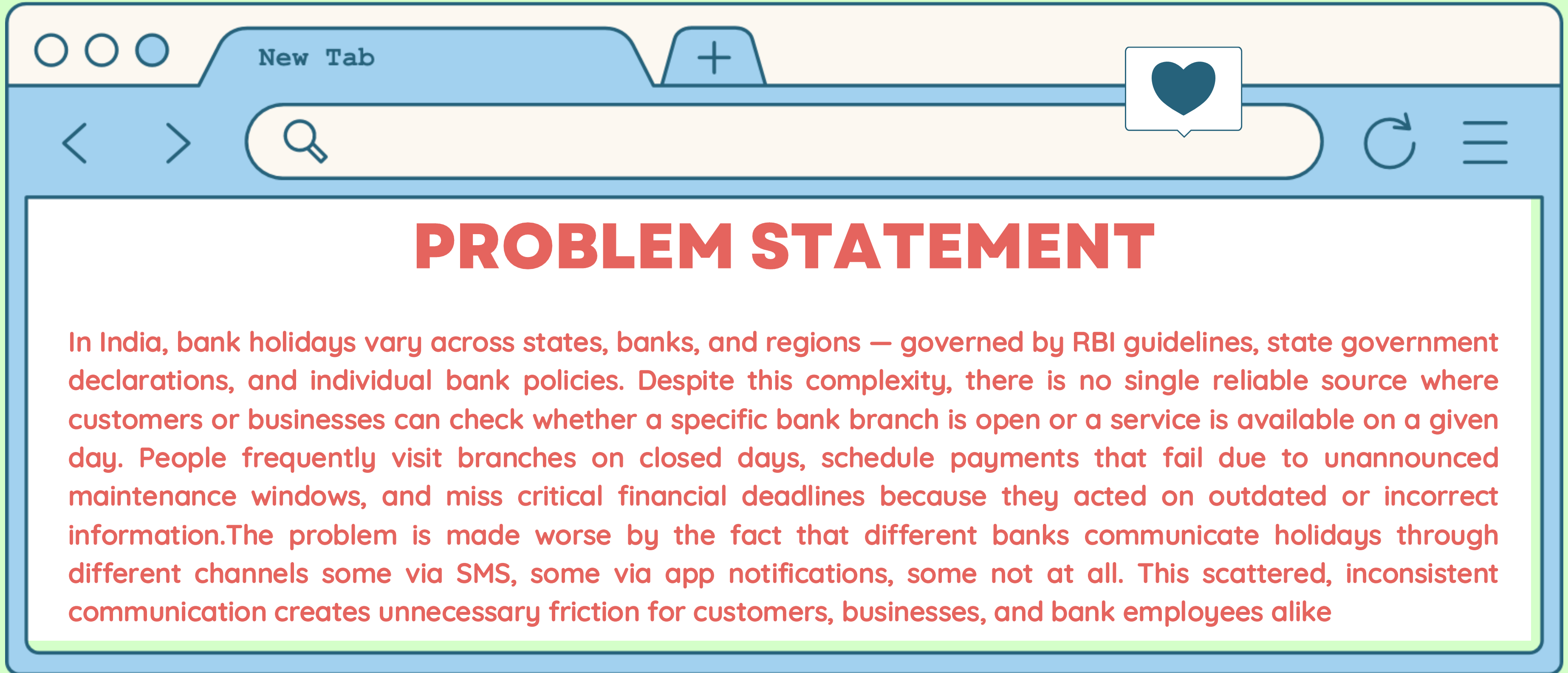
K ANUSRI - 2520030432



INTRODUCTION

A Bank Holiday & Service Calendar is an organized schedule that displays official bank holidays, working days, and the availability of banking services throughout the year. It helps customers, employees, and businesses plan financial activities such as transactions, payments, loans, and customer support efficiently.

This calendar provides awareness about holidays when banks remain closed and highlights essential services like ATM access, online banking, mobile banking, and digital payments that continue to operate. It plays an important role in improving time management, avoiding delays in transactions, and ensuring smooth banking operations for everyone.



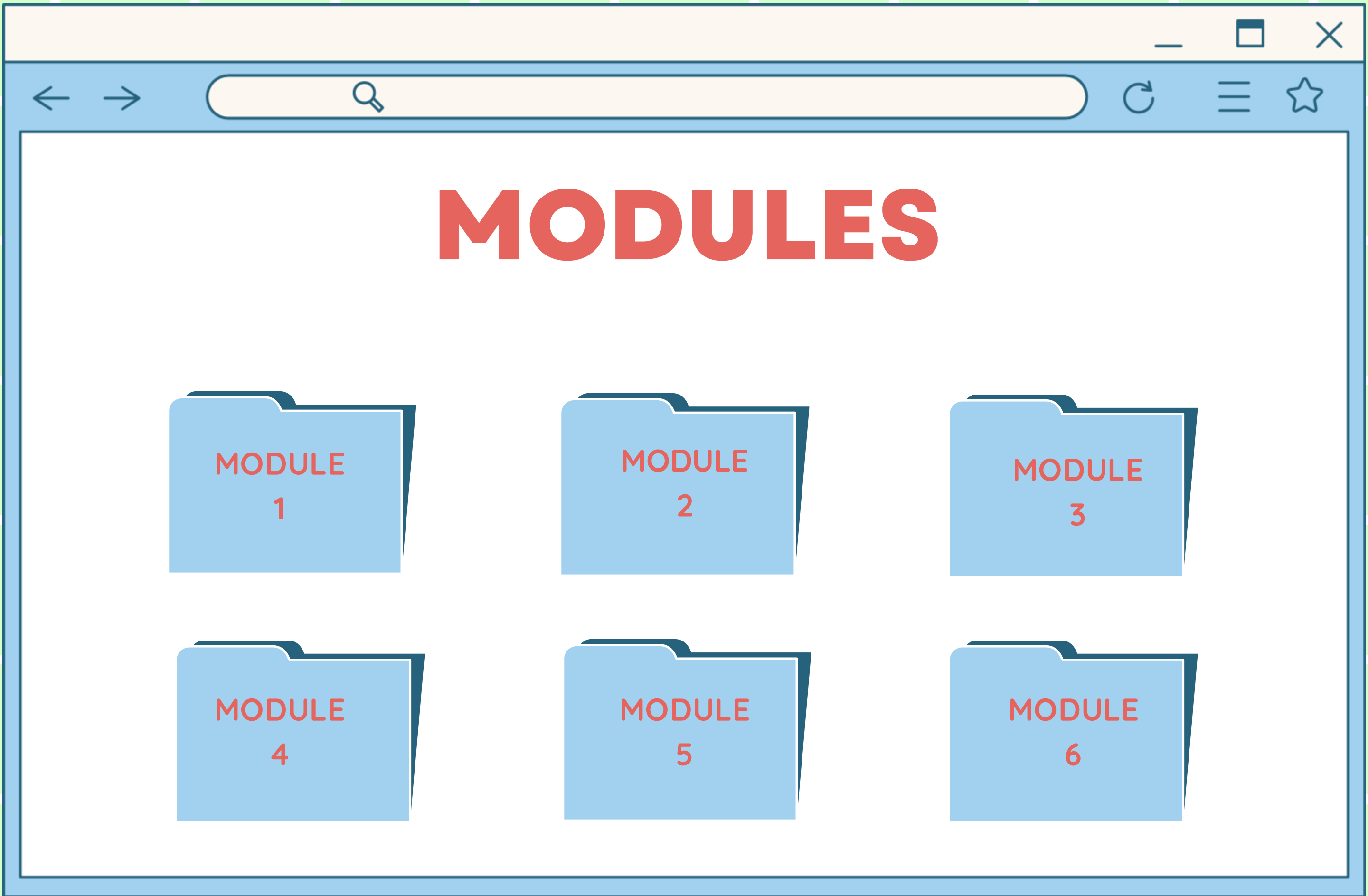


ABSTRACT

The Bank Holiday & Service Calendar is a web application designed to help banks efficiently manage and communicate holiday schedules, service downtimes, and branch operating hours to customers, employees, and businesses — all through a single, centralized digital platform.

Customers can search by date, bank, or region to instantly check whether a branch is open or a service is available. They also receive real-time notifications about upcoming holidays, planned maintenance, and changes in service availability. Administrators are provided a secure dashboard to add, edit, and manage all calendar events and service announcements without requiring technical knowledge.

Key features of the application include a dynamic holiday calendar, region and bank-wise filters. The project significantly improves transparency between banks and their customers, minimizes failed transactions, eliminates unnecessary branch visits, and empowers users to plan their financial activities with confidence and accuracy throughout the year.



MODULE 1 - Foundations of Front-End Engineering & Framework Design

IMPLEMENTATION

- Holiday list
- Region filter

Holiday list

Displays all bank holidays in an organised, date-wise list fetched from the database. The UI updates automatically when holiday data changes, demonstrating declarative rendering.

Region filter

Allows users to filter the holiday list by state or region using a dropdown. Filter state is passed down to the list component, demonstrating unidirectional data flow.

MODULE 2 - JavaScript & TypeScript Engineering for Frameworks

IMPLEMENTATION

- Export
- History

 Export

Lets users download the holiday calendar as a CSV or PDF file using async file generation. Uses TypeScript interfaces to ensure the exported data is correctly typed and structured.

 History

Tracks and displays a log of all changes made to the calendar, such as holidays added or edited. Uses TypeScript generics and closures to maintain a typed, persistent action history.

IMPLEMENTATION

- Add form
- Branch hours

 Add form	A controlled React form that allows admins to add new holidays with fields for date, name, and region. Every input is tied to state via hooks, with validation handled before submission.
 Branch hours	Displays the operating hours of each bank branch using reusable time-slot components. Branch data is passed as props and styled using Tailwind CSS for a consistent layout.

MODULE 4 - State Architecture, Async Data Engineering & API Integration

IMPLEMENTATION

- Service downtime
- Notify customers

 Service downtime	Fetches and displays planned maintenance windows and service outages from a backend API. Handles stale data and async flow to ensure customers always see the most current downtime information.
 Notify customers	Enables admins to send holiday or downtime alerts to customers via email or SMS through an API call. Uses async/await and promise chains to trigger, queue, and confirm delivery reliably.

MODULE 5 - Routing, Forms, Accessibility & Performance Engineering

IMPLEMENTATION

- Admin panel
- ARIA Accessibility

Admin panel

A protected dashboard accessible only to authenticated admins, implemented using a PrivateRoute component in React Router. Admins can manage holidays, branches, and service updates from one place.

ARIA accessibility

Adds ARIA labels, roles, and keyboard navigation support across all interactive elements of the website. Ensures the platform is usable for all users including those relying on screen readers.

MODULE 6 - Build Systems, Testing, CI/CD & Deployment Engineering

IMPLEMENTATION

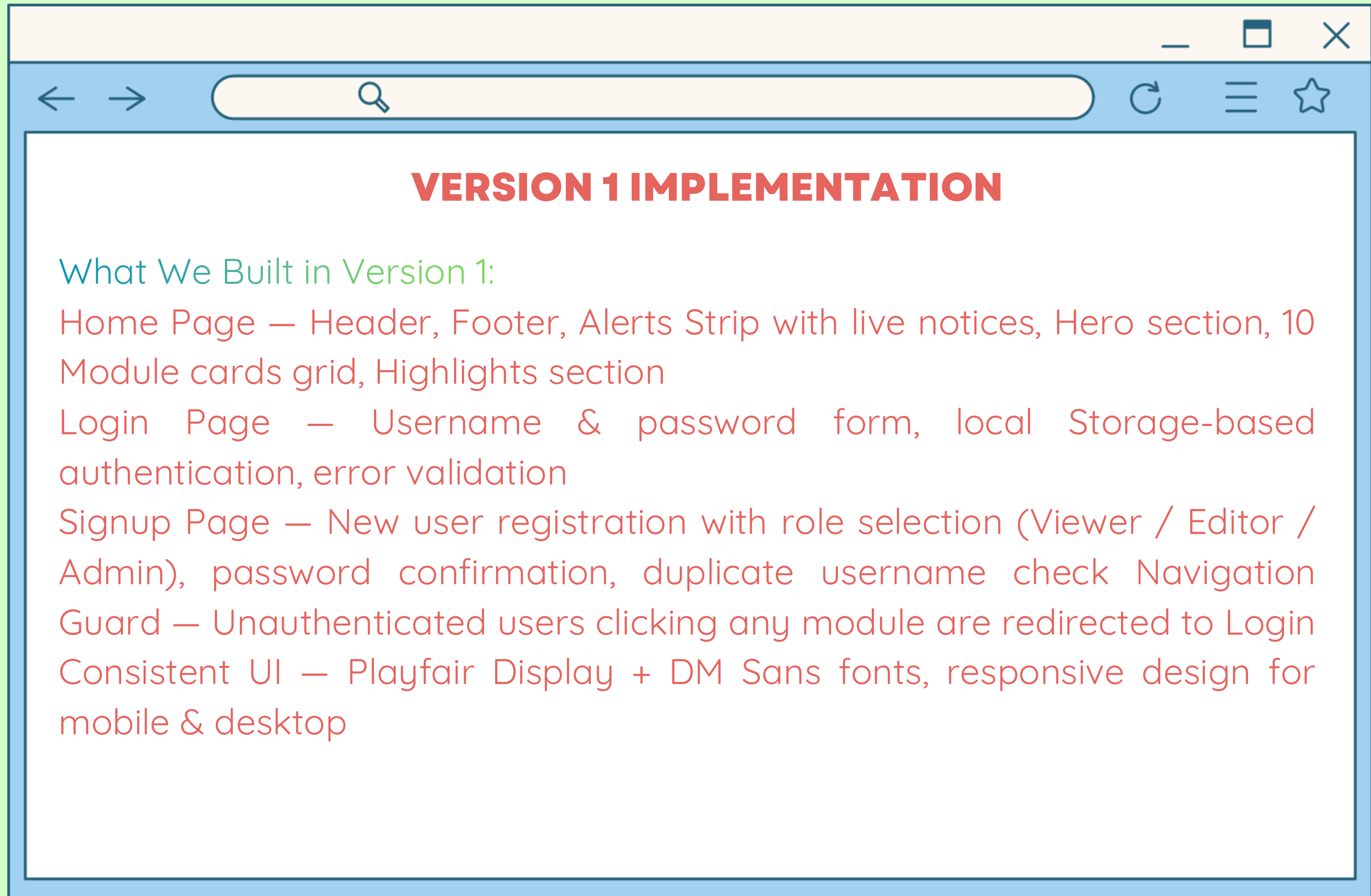
- Notifications
- Deployment

Notifications

An in-app bell icon component that shows unread holiday and service alerts to the logged-in user. Unit tested with Jest and integration tested with React Testing Library to verify correct rendering and dismiss behaviour.

Deployment

The app is built using Vite and deployed to Netlify with automatic CI/CD triggered on every GitHub push. A Lighthouse audit is run on the live site to evaluate performance, accessibility, and best practices scores.



BANK HOLIDAY & SERVICE CALENDER _ V1/

- node_modules/
- public/
- src/
 - assets/
 - hero.png
 - react.svg
 - vite.svg
 - pages/
 - Home.jsx
 - Login.jsx
 - Signup.jsx
 - styles.css
- App.css
- App.jsx
- index.css
- main.jsx
- .gitignore
- eslint.config.js
- index.html
- package-lock.json
- package.json
- README.md
- vite.config.js

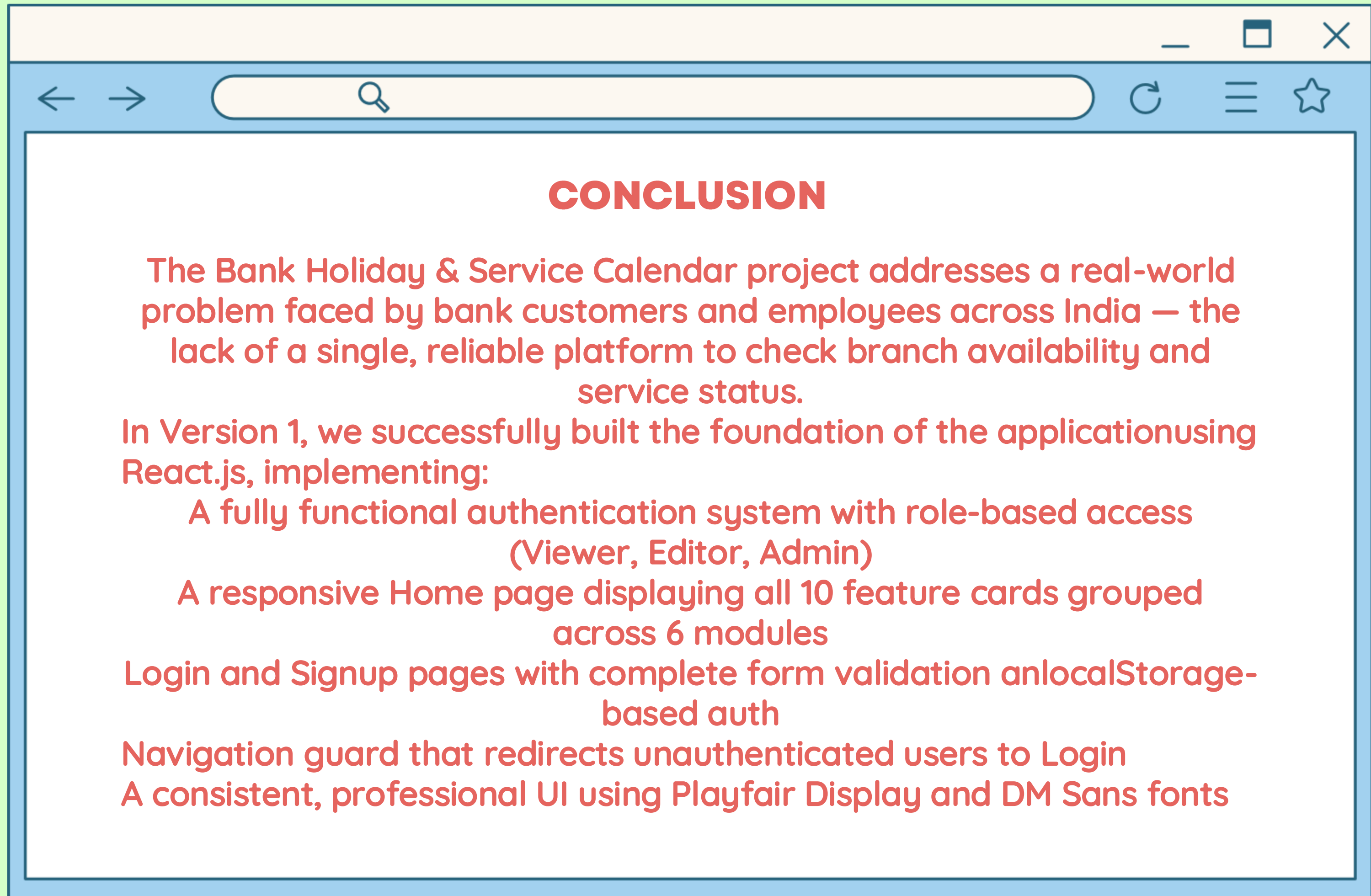
PROJECT STRUCTURE

MAIN EDITED LAYOUT FOR VERSION 1

BANK HOLIDAY & SERVICE CALENDER _ V1/

- src/
 - App.jsx
 - pages/
 - Home.jsx
 - Login.jsx
 - Signup.jsx
 - styles.css

VERSION ROADMAP		
Version	Status	Features
V1	☒ Completed	Home, Login, Signup, Role-based registration, Navigation guard, Responsive UI
V2	☐ Planned	Holiday List page, Add Holiday form, Region filter, Branch Hours page
V3	☐ Planned	Service Downtime, Notify Customers, Export (CSV/PDF), Admin Panel, History





BANK HOLIDAY & SERVICE CALENDAR

VERSION 2



PAGES & FILE STRUCTURE

FILE STRUCTURE

```
src/  
  components/  
    AddHoliday.jsx  
    HolidayList.jsx  
  pages/  
    Home.jsx  
    Login.jsx  
    Signup.jsx  
    # styles.css  
  App.jsx
```

ROUTES

/ Home

Landing page — hero, module cards

/login

Auth form, localStorage session

/Signup

Register with role selection

/ HolidayList

Calendar + list, state filters

/ AddHoliday

Protected — Admin/Editor only



KEY FEATURES — VERSION 2



State-wise Data

- 26 states & UTs · 200+ holidays
- getSeedData() + localStorage merge



Dual View

- List table + Month calendar grid
- Respects state/type/search filters



Role-Based Access

- Admin: add+delete all
- Viewer: read-only



Advanced Filtering

- 26 states · 5 types · text search
- All combined in real time



Add Holiday Form

- Name, Date, Type, Region, Desc
- Dupe check → 1.8s redirect



5 Holiday Types

- National · Bank · Restricted
- Regional · State — color-coded



ROLE-BASED ACCESS & ADDHOLIDAY FLOW



ADMIN

- ✓ View holidays
- ✓ Add holidays
- ✓ Delete any holiday
- ✓ Access AddHoliday



VIEWER

- ✓ View holidays
- ✓ List & calendar only
- ✓ No delete action
- ✓ Blocked from AddHoliday

1

Auth Guard

Not logged in
→ Login screen

2

Role Guard

Viewer/null
→ Access denied

3

Validate

Empty fields
→ inline error

4

Dupe Check

Same name+date
→ reject

5

Save

Append + toast
→ redirect



SUMMARY & VERSION ROADMAP

✓ VERSION 2 DELIVERS

- ✓ 200+ holidays · 26 states & UTs
- ✓ Dual view: List table + Calendar grid
- ✓ 5 color-coded holiday types
- ✓ Role-based access: Admin/Editor/Viewer
- ✓ Add holiday with validation & dupe check
- ✓ Inline delete for Admin/Editor
- ✓ Combined state + type + search filters
- ✓ Breadcrumb nav + protected fallback UI

🗺️ VERSION ROADMAP

V

1



Done

- Home, Login, Signup pages
- Role-based auth (Viewer/Editor/Admin)
- Navigation guard + responsive UI

V

2



Done

- HolidayList + AddHoliday
- State-wise data + Calendar view
- RBAC + delete + advanced filters

FINAL



Planned

- Backend API + JWT auth
- Export CSV/PDF · Notifications
- ARIA accessibility + Deployment



BANK HOLIDAY

& SERVICE CALENDAR

**FINAL
VERSION**



FINAL VERSION — FILE 1

ServiceDowntime.jsx

IMPLEMENTATION

Schedule Maintenance

Admins can schedule planned downtime for systems like Core Banking, ATM Network, NEFT/RTGS, UPI, Mobile App and more with date, start/end time and reason.

Status Management

Each downtime entry moves through lifecycle states: Scheduled → Active → Completed. Admins can update status or delete entries; seed data pre-populates realistic windows.

Audit Integration

Every new downtime entry is auto-written to the auditLog in localStorage, recording the system name, date, acting user, and timestamp for full traceability.



FINAL VERSION — FILE 2

NotificationCentre.jsx

IMPLEMENTATION

Compose & Send Alerts

Admins can compose SMS, Email, or combined alerts targeting All Customers, Business Accounts, Retail, Loan, or Priority Banking segments.

Quick Templates

Three built-in templates — Holiday Closure, System Maintenance, and Branch Hours Change — auto-fill subject and body to speed up notification drafting.

Draft & History

Notifications can be saved as drafts or sent immediately. All entries (Sent / Draft) are stored in localStorage with preview mode and audit log integration.



FINAL VERSION — FILES 5 & 6

AdminPanel.jsx & ExportModule.jsx

Role-Based User Management

Admins view all registered users, edit their roles (User ↔ Admin), and delete accounts. Self-deletion is blocked to prevent lockout.

Stats & Filters

Live counts of total users, admins, and regular users. Filter buttons and search bar narrow the user table instantly.

CSV Export

Downloads all filtered holidays as a comma-separated file, ready for Excel or Google Sheets. Column headers match the portal table.

iCal & PDF Export

iCal (.ics) syncs directly with Google Calendar, Apple Calendar, or Outlook. PDF opens a print-ready browser window formatted for A4/letter output.



FINAL VERSION — FILES 3 & 4

BranchHours.jsx & AuditHistory.jsx

Branch Hours Management

Admins can define and edit operational hours per branch. Seed data covers major cities. Edits are stored in localStorage and written to the audit log.

City-wise Coverage

Tracks opening/closing times and lunch breaks for each branch location. Role-based: only admins can edit; users see a read-only view.

Timestamped Audit Log

Every action across the portal (add holiday, delete, role change, send notification, schedule downtime) is recorded with user and ISO timestamp.

Filter & Search

Admins can filter the log by action type, user, or keyword. Seed data provides 6 realistic entries. Clearing the log requires confirmation.

SUMMARY & VERSION ROADMAP

✓ FINAL VERSION DELIVERS

- ✓ Service Downtime Scheduler (6 systems)
- ✓ Notification Centre with 3 templates
- ✓ Branch Hours management (city-wise)
- ✓ Audit History with 9 action types
- ✓ Admin Panel — User RBAC management
- ✓ Export: CSV, iCal (.ics), PDF/Print
- ✓ Full localStorage persistence across all modules
- ✓ Complete audit trail for every action

📋 VERSION ROADMAP

V1

Done ✓

- Home, Login, Signup
- Role-based auth
- Navigation guard + responsive UI

V2

Done ✓

- HolidayList + AddHoliday
- State-wise data + Calendar view
- RBAC + delete + advanced filters

FINAL

Done ✓

- Downtime, Notifications, Branch Hours
- Audit History + Admin Panel
- Export CSV / iCal / PDF



KEY FEATURES — FINAL VERSION



Downtime Scheduler

Scheduled, Active, Completed & Cancelled statuses



Notification Centre

3 quick templates, SMS/Email/Both, 5 audience types



Branch Hours

City-wise hours management with edit & audit trail



Audit History

9 action types logged with user, timestamp & filters



Admin Panel

User RBAC — edit roles, delete users, live stats



Export Module

CSV, iCal (.ics), and print-to-PDF with state/type filters

WEBPAGE STRUCTURE

Bank Holiday & Service Calendar · React + Vite Application

 Pages (src/pages/)

- Home.jsx — Main dashboard
- Login.jsx — Auth entry point
- Signup.jsx — Registration form
- styles.css — Page-level styles

 App Root (src/)

- App.jsx — Router & layout shell
- App.css — Global component styles
- main.jsx — Vite entry point
- index.css — CSS resets & base

 Config & Build

- vite.config.js — Build & dev server
- eslint.config.js — Linting rules
- index.html — HTML entry template
- package.json / lock — Dependencies

 Components (src/components/)

 AddHoliday.jsx

Admin form to add holidays

 AdminPanel.jsx

Role-based admin dashboard

 AuditHistory.jsx

9-action audit trail log

 BranchHours.jsx

Branch working-hours viewer

 ExportModule.jsx

CSV / iCal / PDF export

 HolidayList.jsx

200+ holidays, dual-view

 NotificationCentre.jsx

SMS & Email templates

 ServiceDowntime.jsx

Downtime status lifecycle



CONCLUSION

The Bank Holiday & Service Calendar project has successfully evolved into a complete, production-ready portal that addresses the real-world challenge of fragmented holiday and service information in Indian banking. Across three versions, every planned module has been fully implemented and integrated.

 Authentication system with role-based access (Admin / User)

 Service downtime scheduler with status lifecycle management

 Branch hours, audit history with 9 action types, admin user control

 17 Holiday calendar with 200+ entries, dual-view, and state filters

 Notification centre with templates and SMS/Email channels

 Export module: CSV, iCal, and print-to-PDF with filters

